CS 5306 / INFO 5306
Fall 2025

Project 3

The goal of this project is to obtain some experience with using the crowdsourcing platform Prolific as a "requester," doing so in a way that gives you some insight into the platform. You will do so in one of two ways: (1) exploring how human perceptual and/or psychological factors impact the effectiveness (accuracy, speed, etc.) of crowd work or (2) exploring how crowd work performance compares to LLMs doing the same task.

OPTION 1:

You read multiple papers that described different ways cognitive biases might impact the effectiveness of a project that uses microlabor. Further papers were discussed in lecture. Your goal would be to select a cognitive bias and see whether you can show how that bias does indeed impact results. Some example papers that might give you inspiration for your project:

- "Priming for Better Performance in Microtask Crowdsourcing Environments" showed that the inclusion of a photo of a laughing baby (irrelevant to the task) or the playing of up-tempo music improves the performance of crowd workers.
- "Influencing Visual Judgment through Affective Priming" showed a similar effect when having workers first read a "positive" short story.
- "Improving Learning through Achievement Priming in Crowdsourced Information Finding Microtasks" demonstrated that showing inspirational quotes in tasks can improve results for many workers.
- "Chain Reactions: The Impact of Order on Microtask Chains" showed that the sequence of tasks that you give a worker can affect worker performance.
- "And Now for Something Completely Different: Improving Crowdsourcing Workflows with Micro-Diversions" showed that "micro-diversions" – such as a brief game, video, or comic – can improve worker performance and retention when working on a series of tasks.
- "Cognitively Inspired Task Design to Improve User Performance on Crowdsourcing Platforms" demonstrated that simple visual layout choices based on perceptual psychology (such as to reduce visual search time or working-memory loads) leads to higher accuracy results.
- "Pay It Backward: Per-Task Payments on Crowdsourcing Platforms Reduce Productivity" showed that different compensation schemes discussed in traditional labor economics lead to differing performance outcomes in crowd work.

These papers often have projects that are more elaborate than you'll need for the purposes of this project. Rather, they provide examples of the kind of thinking the project is targeting.

For this project you should (1) identify some psychological factor that you conjecture might impact the outcomes of microlabor, (2) define a precise experiment that tests it, and then (3) report on your findings. You can replicate a result found in an existing paper (whether part of the course readings or not), or investigate something of your own creation. To do the latter, you can consult the following resources:

- Articles:
  - The Cognitive Bias Cheat Sheet
  - Wikipedia's article on Cognitive Biases, especially the subsection on "Methods"

    - Wikipedia's article on Heuristics (Psychology), especially the subsection on "Informal models of heuristics"
    - Wikipedia's article on Persuasion, especially the subsection on "Methods"
- Lists of cognitive biases:
    - Wikipedia's list of cognitive biases
    - Wikipedia's index of its pages for different cognitive biases

You do not need to use an item from these resources if you've otherwise come across some psychological factor that interests you. These are given to help you explore possibilities.

IMPORTANT: Your goal is NOT to show that the crowdsourcing platform workers exhibit cognitive biases. They're human, so of course they do. The goal is to find some way in which a cognitive bias can impact the outcomes of their work in some fashion.

Think carefully about the task you ask the workers to perform – it needs to be something where answers will differ for the specific phycological factor that you are targeting. Don't over-complicate the task. For many projects the most straightforward task will involve having a worker either estimate some numerical value (like number of dots in an image) or to select a label from some discrete set (like which animal is in an image). You will likely want to pick a task where you know the correct answer(s), so you can assess the performance of workers.

As with past projects you do not need to be correct in your prediction. You will be graded on the thoughtfulness and clarity of your question and experiment(s).

OPTION 2

You read papers that described comparisons of LLMs with crowdsourced labor. Further papers were discussed in lecture. Your goal would be to propose some interesting hypothesis about how LLMs might perform on some task compared to humans and see the extent to which LLMs perform comparably to or better than people on some tasks. Refer to the papers you read and discussed in class for ideas about possible projects to pursue.

Deliverables:

- Proposal:
  Due: Friday, November 13, 11:59pm.
  It should have the following sections in the following order:
    - Title
    - Names and NetIDs of teammates
    - A brief description of the psychological factor. This should not be a lengthy treatment, it should be one simple paragraph giving enough of a sense of the factor to ground the rest of the report. This section has nothing to do with crowdsourcing or human computation – you are simply describing a psychological characteristic of people. This section can include a link to a good overview of the topic (Wikipedia or other websites are fine).
    - Explain what effect that you think the psychological factor will have on worker performance.
    - Describe the exact experiment you plan to conduct. Make sure to highlight
        - the question you are testing and the answer you are expecting,
        - the tasks that you will give workers, and
        - the source of data for the tasks (if relevant).

- One-slide summary:
  Due: Sunday, December 6, 10:00pm
  Because the final report is not due until December 15, this will necessarily be a summary of what you're working on, not your final outcome. Explain what you're investigating and how you're investigating it.

- In-class presentation:
  Due: Monday, December 7, 10:00am.
  This is your one-minute presentation on your project using the slide you previously submitted.

- Final report:
  Due: Tuesday, December 15, 9:00am.
  Your final submission will once again be a report of roughly 5 pages. Go longer if you need to, especially if this involves the inclusion of images or graphs. Your report should have the following sections in the following order:
    - Title
    - Names and NetIDs of teammates
    - A brief description of the question you're asking and what hypothesized result you were expecting.
    - Describe the exact experiments you conducted. Make sure to highlight
      a. the precise question you tested,
      b. the specific tasks that you gave workers, and
      c. the source of data for the tasks (if relevant).
    - A discussion of your outcomes.